Introduction to Machine Learning
Fall 2025
University of Science and Technology of China

Lecturer: Zhihui Li, Xiaojun Chang — Homework 3
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Notice, to get the full credits, please present your solutions step by step.

Exercise 1: Affine Sets and Projections

(1) Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and consider

$$U = \{x \in \mathbb{R}^n : Ax = b\}.$$

(a) Show that U is an affine set. Identify its direction space in terms of A.

(b) Prove that for any $x_0 \in U$,

$$U = x_0 + \mathcal{N}(A),$$

where $\mathcal{N}(A)$ denotes the null space of A.

(2) Suppose $A \in \mathbb{R}^{m\times n}$ has full row rank and $U = \{x : Ax = b\}$ is nonempty.

(a) Consider the projection of a point $y \in \mathbb{R}^n$ onto U:

$$p = \mathop{\mathbf{argmin}}_{x\in U} \|x - y\|_2^2.$$

Use Lagrange multipliers to derive the formula

$$p = y - A^\top(AA^\top)^{-1}(Ay - b).$$

(b) In $\mathbb{R}^3$, let

$$U = \{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 3\}, \quad y = (1, 0, 0)^\top.$$

Compute explicitly the projection p of y onto U.

Solution 1: ■

Exercise 2: Convex Sets, Images and Preimages

Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and let $A \in \mathbb{R}^{m\times n}$, $a \in \mathbb{R}^m$.

(1) Show that the affine image of C,

$$D = \{y \in \mathbb{R}^m : y = Ax + a,\, x \in C\},$$

is convex.

(2) Define the affine preimage of C by

$$E = \{y \in \mathbb{R}^m : Ay + a \in C\}.$$

Show that E is convex.

(3) (Epigraph and hypograph)

(a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. The epigraph of f is

$$\textbf{epi}\ f = \{(x,t) \in \mathbb{R}^{n+1} : t \geq f(x)\}.$$

Show that **epi** f is a convex set.

(b) Let $g : \mathbb{R}^n \to \mathbb{R}$ be concave. Its hypograph is

$$\operatorname{hypo} g = \{(x,t) \in \mathbb{R}^{n+1} : t \leq g(x)\}.$$

Show that hypo g is convex.

(4) (Interior and relative interior – computation) You do not need to prove your answers, but you should explain them briefly.

(a) Find **int** C and **relint** C for

$$C = \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 < 4,\ x_3 \geq 0\}.$$

(b) Let

$$K = \{x \in \mathbb{R}^n : x_i \geq 0,\ \sum_{i=1}^{n} x_i = 1\}$$

be the standard probability simplex. Determine **int** K (as a subset of $\mathbb{R}^n$) and **relint** K.

Solution 2: ■

Exercise 3: Cones and Farkas-type Results

Let $A = (a_1, \dots, a_n) \in \mathbb{R}^{m \times n}$ and consider the conic hull

$$\operatorname{cone}(A) = \left\{ \sum_{i=1}^{n} \alpha_i a_i : \alpha_i \geq 0 \right\} \subseteq \mathbb{R}^m.$$

(1) Show that cone(A) is a convex cone, i.e.,

$$x, y \in \operatorname{cone}(A),\ \ \lambda, \mu \geq 0 \ \Rightarrow\ \lambda x + \mu y \in \operatorname{cone}(A).$$

(2) Show that cone(A) is closed.

(3) (Gordan's theorem) Let $A \in \mathbb{R}^{m \times n}$. Prove that exactly one of the following two systems has a solution:

- There exists $x \in \mathbb{R}^n$ such that $Ax > 0$ (componentwise).
- There exists $y \in \mathbb{R}^m$, $y \geq 0$, $y \neq 0$, such that $A^\top y = 0$.

Hint: Consider appropriate cones and use separation theorems (you may assume standard separation results for closed convex cones).

(4) Show how Farkas' Lemma can be deduced from Gordan's theorem in part (3).

Solution 3: ■